

Installation Guide

ARCAD Transformer

Microservices

Version 24.2



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Table 1: Contact ARCAD

Preface

Document purpose

This document is intended to guide users through the installation process for the following ARCAD Transformer Microservices module(s):

- The ARCAD Transformer Microservices Server
- The ARCAD Transformer Microservices plug-in for RDi

Intended audience

This document is intended for system administrators, ARCAD administrators, or any person in charge of installing ARCAD products.

Publication record

Unless stated otherwise, all content is valid for the most current version of ARCAD Transformer Microservices listed as well as every subsequent version.

Product version	Document version	Publication Date	Update record
≥ 24.2	1.1	September, 2024	No functional changes • Updated the installation procedure for the ARCAD Transformer Microservices plug-in • Updated the system requirements for RDi
24.1	1.0	June, 2024	Initial publication.

Table 2: ARCAD Transformer Microservices Installation Guide publication record

Related documentation

ARCAD technical documentation can be accessed from the product's online help or by logging into the [Customer Portal](#) on our website.

ARCAD Transformer Microservices User Guide.

Table 3: Related documentation

 **Important!**

The **User Guide** documentation for ARCAD Transformer Microservices is now available from the ARCAD Transformer Microservices Server when it is installed and started, by using the following URL:

WEB : [IP]:[port]/docs/index.html.

By default, the [port] is 5265.

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1 About ARCAD Transformer Microservices

ARCAD Transformer Microservices is an integral tool in the process of modernizing legacy applications by honing in on specific sections of code that possess unique characteristics.

The emphasis on uniqueness in code sections allows for a gradual evolution of application architecture from monolithic to modular, paving the way for the implementation of highly relevant architectures like microservices. Adopting a microservices architecture offers a myriad of benefits to both developers and enterprises:

For developers, the implementation of microservices architecture offers several significant advantages.

By avoiding the accumulation of a large code base, it simplifies the maintenance and the addition of new features. It also enhances deployment processes and reduces IDE load times.

Additionally, debugging becomes more straightforward, and tracking code dependencies is less cumbersome, facilitating smoother project workflows.

As for enterprises, it enables more frequent deliveries and shorter delivery times, allowing faster feedback cycles.

It also enhances resource management and improves service availability, ultimately leading to a better user experience. Additionally, identifying and eliminating duplicate services can significantly reduce development expenses and operational management costs.

ARCAD Transformer Microservices is designed to facilitate the implementation of such architecture by providing functionalities that support the generation of microservices based on IBM i applications.

To achieve this goal, the product focuses on:

- auditing the risk associated with externalizing a selected portion of code before centralizing it,
- determining where a selected piece of code is used within an application, and
- ensuring the uniqueness of the call to centralized code within the application.



SYSTEM REQUIREMENTS

2 System requirements for the ARCAD Transformer Microservices Server

2.1 Windows

2.1.1 Hardware requirements

The following table lists the hardware components that are required for the ARCAD Transformer Microservices Server on Windows x64.

Requirement	Value
System Architecture	64 bit processor
CPU	DUAL Core processor minimum
Physical memory (RAM)	4GB minimum, 8GB recommended ¹
Hard disk	RAID storage or any secured storage with redundancy, backup and monitoring.
Disk Space	Typical Install Type total: 10GB ²
Network	Any TCP/IP capable network interface

Table 4: ARCAD Transformer Microservices Server Hardware Requirements - Windows

2.1.2 Software requirements

The following table lists the software requirements for the ARCAD Transformer Microservices Server on Windows x64.

Requirement	Value
Operating system	Any Microsoft supported server OS with Java (eg windows server 2016 any version)
Network protocol	• TCP/IPv4 • TCP/IPv4 with SSL
Filesystem	NTFS or any journalized filesystem

Table 5: ARCAD Transformer Microservices Server Software Requirements - Windows

¹Physical memory is used by this entity and all subsequent processes, operating systems and any other software installed on the server. Please refer to additional software vendors for additional physical memory requirements.

²This size can be higher depending on the options selected and typical usage of the software, refer to Hard Disk Space requirement planning section to determine your needs.

2.1.3 Remote Desktop Services

ARCAD Microservices supports installing, configuring, and running all of its software through Remote Desktop Services, formerly known as Terminal Services, on Windows. To install, it is recommended that you start all configuration tools from the Terminal Server console session of the server using **mstsc** or console.

2.1.4 Host name resolution

Typically, the partition on which you want to install the ARCAD Transformer Microservices Server is connected to a network. Ensure that the partition host name is resolvable through a Domain Name System (DNS) or the centrally-maintained TCP/IP host table (go CFGTCP, option 10).

Use the PING command to ensure that your computer host name is resolvable.

Example

```
PING RMTSYS(hostname)
Verifying connection to host system
hostname.domain.com at address 192.168.0.10.
PING reply 1 from 192.168.0.10 took 18 ms. 256 bytes. TTL 64.
```

If your computer host name does not resolve, contact your system administrator.

2.1.5 Users, Groups and Environments for the ARCAD Microservices Server

Note

Please refer to your OS and system administrator for common security practices.

The environment must be clean of variables which might affect java software such as *JAVA_HOME*, *JDK_HOME*, *JRE_HOME*, *JAVA_OPTIONS*, *JAVA_OPTS*, *JVM_OPTS*, *CLASSPATH*, etc.

Important!

PATH must contain all java related binaries and tools.

TMP, TEMP, TMPDIR and TEMPDIR can be set to any desired temporary storage location.

2.2 Linux

2.2.1 Hardware requirements

The following table lists the hardware components that are required for the ARCAD Transformer Microservices Server on Linux x64.

Requirement	Value
System Architecture	64 bits processor

Table 6: ARCAD Transformer Microservices Server Hardware Requirements - Linux

Requirement	Value
CPU	DUAL Core processor minimum
Physical memory (RAM)	4GB minimum, 8GB recommended ¹
Hard disk	RAID storage or any secured storage with redundancy, backup and monitoring
Disk Space	Typical Install Type total: 10GB ²
Network	Any TCP/IP capable network interface

Table 6: ARCAD Transformer Microservices Server Hardware Requirements - Linux

2.2.2 Software requirements

The following table lists the software requirements for the ARCAD Transformer Microservices Server on Linux x64.

Requirement	Value
Operating system	Linux 64 bits
Java JRE	Java runtime 8; 64 bits required
Runlevel	Server should be started in multi-user mode, commonly runlevel 3 or 5
Network protocol	- TCP/IPv4 - TCP/IPv4 with SSL
Filesystem	Ext3, ext4 or any journalized filesystem

Table 7: ARCAD Transformer Microservices Server Software Requirements - Linux

2.2.3 Host name resolution

Typically, the partition on which you want to install the ARCAD Transformer Microservices Server is connected to a network. Ensure that the partition host name is resolvable through a Domain Name System (DNS) or the centrally-maintained TCP/IP host table (go CFGTCP, option 10).

Use the *PING* command to ensure that your computer host name is resolvable.

Example

```
PING RMTSYS(hostname)
Verifying connection to host system
hostname.domain.com at address 192.168.0.10.
PING reply 1 from 192.168.0.10 took 18 ms. 256 bytes. TTL 64.
```

If your computer host name does not resolve, contact your system administrator.

¹Physical memory is used by this entity and all subsequent processes, operating systems and any other software installed on the server. Please refer to additional software vendors for additional physical memory requirements.

²This size can be higher depending on the options selected and typical usage of the software, refer to Hard Disk Space requirement planning section to determine your needs.

2.2.4 Users, Groups and Environments for the ARCAD Microservices Server

Note

Please refer to your OS and system administrator for common security practices.

The environment must be clean of variables which might affect java software such as `JAVA_HOME`, `JDK_HOME`, `JRE_HOME`, `JAVA_OPTIONS`, `JAVA_OPTS`, `JVM_OPTS`, `CLASSPATH`, etc.

Important!

PATH must contain all java related binaries and tools.

`TMP`, `TEMP`, `TMPDIR` and `TEMPDIR` can be set to any desired temporary storage location.

2.3 IBM i

2.3.1 Hardware requirements

The following table lists the hardware components that are required for the ARCAD Transformer Microservices Server on IBM i.

Requirement	Value
Physical memory (RAM)	4GB minimum, 8GB recommended ¹
Disk Space	Typical Install Type total: 10GB ²
Network	Any TCP/IP capable network interface

Table 8: ARCAD Transformer Microservices Server Hardware Requirements - IBM i

2.3.2 Software requirements

The following table lists the software requirements for the ARCAD Transformer Microservices Server on IBM i.

Requirement	Value
Operating system	IBM i ≥ 7.1 or above
Network protocol	- TCP/IPv4 - TCP/IPv4 with SSL
Java JRE	Java runtime 8; 64 bits required
REXEC	At least one instance of the Remote Execution Server (REXEC) must be running.

Table 9: ARCAD Transformer Microservices Server Software Requirements - IBM i

¹Physical memory is used by this entity and all subsequent processes, operating systems and any other software installed on the server. Please refer to additional software vendors for additional physical memory requirements.

²This size can be higher depending on the options selected and typical usage of the software, refer to Hard Disk Space requirement planning section to determine your needs.

2.3.3 Host name resolution

Typically, the partition on which you want to install the ARCAD Transformer Microservices Server is connected to a network. Ensure that the partition host name is resolvable through a Domain Name System (DNS) or the centrally-maintained TCP/IP host table (go CFGTCP, option 10).

Use the *PING* command to ensure that your computer host name is resolvable.

Example

```
PING RMTSYS(hostname)
Verifying connection to host system
hostname.domain.com at address 192.168.0.10.
PING reply 1 from 192.168.0.10 took 18 ms. 256 bytes. TTL 64.
```

If your computer host name does not resolve, contact your system administrator.



INSTALLATION

3 Installing the ARCAD Transformer Microservices Server

The Microservices Server centralizes all the necessary elements to carry out Microservices processes. The server is the core of ARCAD Transformer Microservices because it is required to use the plug-ins.

Important!

Before installing or updating ARCAD Transformer Microservices, first read carefully the Release Notes.

In these release notes, you will find all the information specific to each release of ARCAD Transformer Microservices, especially specific warnings you should be aware of before installing or upgrading the product (compatibility breaks, exceptional migration procedures, etc.).

3.1 Prerequisites

Reference

For complete details about the technical prerequisites for ARCAD Transformer Microservices, refer to the [System requirements on page 8](#).

3.2 Windows

The ARCAD Transformer Microservices Server runs as a Windows service.

3.2.1 Install

The installation process takes approximately 5 minutes. It is a simple procedure similar to most other Windows applications. For trouble-free installation, it is recommended that you close all active Windows applications before beginning the installation.

Note

If the tool is already installed on your computer, reinstalling it will update it to the new version automatically.

Follow the subsequent steps to install the ARCAD Transformer Microservices Server on Windows.

Step 1 Copy the .exe file to your machine and execute it to launch the wizard. Your profile must have administrator privileges to run the execution file.

Step 2 From the **Select Setup Language** window, select the language for the installation.

Step 3 Review and accept the license agreement.

The **License Agreement** page presents the ARCAD license agreement for you to review. Please read it carefully. When you have reviewed the agreement, select **I accept the agreement**.

Step 4 Confirm the installation location.

The **Select Destination Location** screen displays the default location where ARCAD elements will be installed. If you prefer to install the software elements in a different location, either type in the location, or click **Browse...** to navigate to and select the alternate location.

The default root location is *C:\Program files\ARCAD Solutions*.

A sub-folder for each ARCAD application you install will be created at this location.

 Warning!

The access rights to this folder can then be restricted to the user used to run the Server's Windows service.

Step 5 Install.

The **Ready to Install** screen enables you to review and change or confirm the setup parameters provided and to launch the installation.

To change a setup parameter, click **< Back** and return to the necessary screen.

If you agree with the installation parameters displayed, click **Install** to start the copy phase of the installation process. The process may take a moment while the windows services are launched.

As the installation proceeds, a status bar displays its progress. Each element copied appears above the status bar. Click **Cancel** to interrupt the installation.

Step 6 Define the TCP ports to use.

 Important!

The ports defined here will be used to configure the connection.

Press **Enter/Next** to use the default ports.

The ARCAD Transformer Microservices Server is an independent AFS server with a DB2 for i database managed by ARCAD.

The ARCAD Transformer Microservices default HTTP port is 5265, and the default HTTPS port is 52650.

To disable a port, enter 0 (zero) when asked for the port number.

 Important!

If you are using other ARCAD Group products on the same host, it is recommended to use the default ports to avoid any conflicts with other servers.

If an application is already using it, the default port number is incremented until a free port number is found. It is possible to manually change the port after installation.

Step 7 Select the Windows user account.

The ARCAD Transformer Microservices Server runs as a Windows service. Select the user account that will be used to manage the service.

By default, the local system account is selected. If you do not want to use that account, you can select any user account with enough privileges to run a Windows service, and have a read and write access to the folder where ARCAD Transformer Microservices is installed.

Step 8 Complete the setup process. The final page of the wizard displays confirmation that the setup was a success. Click **Finish** to close the setup wizard.

Step 9 Verify the Windows Service is running.

The final screen of the installation process enables you to automatically open the Local Services Management Console. If the **Open Windows service** checkbox is selected, the **Services** window opens after clicking **Finish**. This window enables you to verify that its status is set to "Started".

Result The ARCAD Transformer Microservices Server is installed and available for use.

3.2.2 Update

When updating the server, the folders containing your configuration and files are kept. However, some personal configuration may be lost during the update. It is recommended to create a backup of your files before updating the server.

The following items are not impacted by the update:

- Database
- Drivers
- Files

Warning!

Uninstalling the ARCAD Transformer Microservices Server will remove all your data. Uninstalling and installing the new version of the server will overwrite the folders containing your configuration and files.

Updating the ARCAD Transformer Microservices Server is similar to the installation process in that it is launched via the installation .exe.

Note

If you are updating without uninstalling, the majority of the installation preferences are saved and automatically reused.

To change the saved values, uninstall completely and define new values by reinstalling from scratch.

Follow the subsequent steps to update the ARCAD Transformer Microservices Server on Windows.

Step 1 Copy the .exe file to your machine and execute it to launch the wizard. Your profile must have administrator privileges to run the execution file.

Step 2 From the **Select Setup Language** window, select the language for the installation.

Step 3 Review and accept the license agreement.

The **License Agreement** page presents the ARCAD license agreement for you to review. Please read it carefully. When you have reviewed the agreement, select **I accept the agreement**.

Step 4 Install.

The **Ready to Install** screen enables you to review and change or confirm the setup parameters provided and to launch the installation.

To change a setup parameter, click **< Back** and return to the necessary screen.

If you agree with the installation parameters displayed, click **Install** to start the copy phase of the installation process. The process may take a moment while the windows services are launched.

As the installation proceeds, a status bar displays its progress. Each element copied appears above the status bar. Click **Cancel** to interrupt the installation.

Step 5 Complete the setup process. The final page of the wizard displays confirmation that the setup was a success. Click **Finish** to close the setup wizard.

Step 6 Verify the Windows Service is running.

The final screen of the installation process enables you to automatically open the Local Services Management Console. If the **Open Windows service** checkbox is selected, the **Services** window opens after clicking **Finish**. This window enables you to verify that its status is set to "Started".

Result The ARCAD Transformer Microservices Server is updated and available for use.

3.2.3 Uninstall

Follow the subsequent steps to uninstall the ARCAD Transformer Microservices Server from Windows.

Step 1 Log into the system with an administrator account.

Step 2 Either launch the *uninstall.exe* located in the installation directory

- or -

Open **Add or Remove Programs** (*Start > Control Panel*), find the module in the list of installed software and select **Uninstall**.

Step 3 Remove any remaining files in the installation path and remove any remaining configuration files in the user directory.

Result The ARCAD Transformer Microservices Server is completely uninstalled.

3.3 Linux

3.3.1 Install

Follow the subsequent steps to install the ARCAD Transformer Microservices Server on Linux.

Step 1 Transfer the *.tar.gz* to the */tmp* directory on the target system.

Step 2 Open a terminal session and go to the installation directory.

 **Example**
`cd /opt`

Note

By default, the server is intended to be installed in the `/opt` directory. If you install it in a different directory, you will have to update the server's scripts (see [Step 7 Edit the scripts.](#)).

Step 3 Execute the following command to extract the `tar.gz` archive into the current directory:

```
tar xzf Microservices-Server-[version number]_Linux.tar.gz
```

Step 4 [Optional] If you need to change the listening port(s) for the server, you must do so now.

The ARCAD Transformer Microservices Server is an independent AFS server with a DB2 for i database managed by ARCAD.

The ARCAD Transformer Microservices default HTTP port is 5265, and the default HTTPS port is 52650.

To disable a port, enter 0 (zero) when asked for the port number.

Important!

If you are using other ARCAD Group products on the same host, it is recommended to use the default ports to avoid any conflicts with other servers.

To change the ports, open the `configuration/com.arcadsoftware.server.restful.cfg` in a text editor and change the following properties:

- **port**=<new HTTP port number> -or- <0> (zero) to disable the HTTP port
- **portssl**=<new HTTPS port number> -or- <0> (zero) to disable the HTTPS port

Step 5 Create a specific user and group to run the server's process with the command:

```
adduser --system --no-create-home --group name
```

Step 6 Change the ownership of the installation directory to give it to the user and group that will run the ARCAD Transformer Microservices Server:

```
chown username:groupname -R /opt/Microservices-Server
```

Step 7 Edit the scripts.

Open the following scripts in a text editor to set the variables to match your execution environment:

- `bin/Microservices-Server`
 - Line 8: set **AFS_HOME** to the installation directory if it was not installed in `/opt/...` by default.
 - Line 12: set **AFS_USER** to the new user.
- `bin/Microservices-Server.service`
 - Lines 8-14: update the paths so they match the installation directory.
 - Line 16: set the user to the new user.
 - Line 17: set the group to be the new user's or any group of your choice.
- configuration files in the `configuration` folder

Step 8 [Optional] Execute the following commands to install the ARCAD Transformer Microservices Server as a systemd service:

```
cd /etc/systemd/system
systemctl link /opt/Microservices-Server/bin/Microservices-Server.service
systemctl enable Microservices-Server
systemctl start Microservices-Server
```

Result The ARCAD Transformer Microservices Server is installed and available for use.

Note

The ARCAD Transformer Microservices Server embeds, in the Linux packaging, the runtime of an open-source version of Java (OpenJDK 1.8). Depending on the needs of the host machine, this runtime can be replaced by an equivalent version managed by the operating system. To do so, delete the **jre** folder located in the server installation folder.

3.3.2 Update

When updating the server, the folders containing your configuration and files are kept. However, some personal configuration may be lost during the update. It is recommended to create a backup of your files before updating the server.

The following items are not impacted by the update:

- Database
- Drivers
- Files

Warning!

Uninstalling the ARCAD Transformer Microservices Server will remove all your data. Uninstalling and installing the new version of the server will overwrite the folders containing your configuration and files.

Follow the subsequent steps to update the ARCAD Transformer Microservices Server on Linux.

Step 1 Transfer the updated `Microservices-Server.tar.gz` and the `Microservices-Server-[version number]_Linux_Update.sh` script to the `/tmp` directory on the target system.

Step 2 Open a terminal session and go into the `/tmp` directory: `cd /tmp`

Step 3 Mark the `_Update.sh` as being executable:

```
chmod +x Microservices-Server-[version number]_Linux_Update.sh
```

Step 4 Open the `_Update.sh` with a text editor and make the following changes:

- Line 9: uncomment and set the **AFS_HOME** variable to the installation directory.
- Line 10: change the **AFS_URL** to match the address and port the server is listening on.
- Line 12: change the **AFS_OSUSER** to the user running the server.
- Line 13: change the **AFS_OSGROUP** to the group running the server.

- Lines 16-19: uncomment the line corresponding to the way the server must be started.
- Lines 22-25: uncomment the line corresponding to the way the server must be stopped.

Step 5 Run the `_Update.sh`:

```
./Microservices-Server-[version number]_Linux_Update.sh
```

Note

You can keep the `_Update.sh` for future updates, unless stated otherwise in the release notes.

Result The ARCAD Transformer Microservices Server is updated and available for use.

3.3.3 Uninstall

Follow the subsequent steps to uninstall the ARCAD Transformer Microservices Server on Linux.

Step 1 Stop the service using the service manager of your Linux distribution or use the script provided.



Example

Service command:

```
systemctl stop Microservices-Server
```

Script command:

```
/opt/Microservices-Server/bin/Microservices-Server stop
```

Step 2 Uninstall the service, if any, using the tools from your Linux distribution.



Example

```
systemctl disable Microservices-Server  
rm /etc/systemd/system/Microservices-Server.service  
systemctl daemon-reload  
systemctl reset-failed
```

Step 3 Remove the software from the installation directory.



Example

```
rm -rf /opt/Microservices-Server
```

Step 4 Remove any remaining files based on the server's configuration.

Result The ARCAD Transformer Microservices Server is completely uninstalled.

3.4 IBM i

The ARCAD Transformer Microservices Server can be installed and updated manually on IBM i or, depending on your security policy, remotely from any machine running Java.

Note

If the tool is already installed on your computer, reinstalling it will update it to the new version automatically.

3.4.1 Install

Follow the subsequent steps to install the ARCAD Transformer Microservices Server on IBM i.

Step 1 Copy the installation *.jar* file to any directory (such as the */tmp*) on the target IBM i IFS or the machine that will orchestrate the remote installation.

Step 2 Launch the installation setup.

If you are installing manually on IBM i:

1. Open a session on the target IBM i using the QSECOFR profile or an equivalent.
2. Open a command line interpreter using the command QSH. Reach the location where you have copied your installation file.
3. In the command line interpreter, launch the setup using the installation command.

**Example**

```
java -jar Setup_Microservices-Server-x.x.x._IBM i.jar
```

If you are installing remotely from Windows or Linux:

1. Launch the setup from the location on the machine orchestrating the installation using the command above. This will open a prompt asking for the **Remote IBM i name/address**.
2. Enter the name/address of the target IBM i.
3. Login using the QSECOFR profile or an equivalent.

Step 3 Define the **AFS Starter Installation library** and iASP which will contain the AFS Starter (a utility program used to start the ARCAD Transformer Microservices Server on IBM i).

By default they are **AFSSTARTER** and ***SYSBAS**.

Step 4 Define the installation location.

**Example**

```
/HOME/...
```

Step 5 Define the TCP ports to use.

**Important!**

The ports defined here will be used to configure the connection.

Press **Enter/Next** to use the default ports.

The ARCAD Transformer Microservices Server is an independent AFS server with a DB2 for i database managed by ARCAD.

The ARCAD Transformer Microservices default HTTP port is 5265, and the default HTTPS port is 52650.

To disable a port, enter 0 (zero) when asked for the port number.

 Important!

If you are using other ARCAD Group products on the same host, it is recommended to use the default ports to avoid any conflicts with other servers.

If an application is already using it, the default port number is incremented until a free port number is found. It is possible to manually change the port after installation.

Step 6 Define the **Job user** that will run the ARCAD Transformer Microservices Server.

Step 7 Define the job queue library (***LIBL**) then the job queue (**JOBQ**) in which the job will be submitted.

 Example

Job queue library [ARCAD_SYS]
Job queue [ARCAD_CTL]

Step 8 Define the name of the AFS instance to register in the AFSSTARTER.

By default it is **PRODUCT-AFS-ID**.

Result The ARCAD Transformer Microservices Server is installed and available for use.

3.4.2 Update

When updating the server, the folders containing your configuration and files are kept. However, some personal configuration may be lost during the update. It is recommended to create a backup of your files before updating the server.

The following items are not impacted by the update:

- Database
- Drivers
- Files

 Warning!

Uninstalling the ARCAD Transformer Microservices Server will remove all your data. Uninstalling and installing the new version of the server will overwrite the folders containing your configuration and files.

Updating the ARCAD Transformer Microservices Server is similar to the installation process in that it is launched via the installation *.jar* file.

 Note

If you are updating without uninstalling, the majority of the installation preferences are saved and automatically reused.

To change the saved values, uninstall completely and define new values by reinstalling from scratch.

Follow the subsequent steps to update the ARCAD Transformer Microservices Server on IBM i.

Step 1 Copy the installation *.jar* file to any directory (such as the */tmp*) on the target IBM i IFS or the machine that will orchestrate the remote installation.

Step 2 Launch the installation setup.

If you are installing manually on IBM i:

1. Open a session on the target IBM i using the QSECOFR profile or an equivalent.
2. Open a command line interpreter using the command QSH. Reach the location where you have copied your installation file.
3. In the command line interpreter, launch the setup using the installation command.

**Example**

```
java -jar Setup_Microservices-Server-x.x.x._IBM i.jar
```

If you are installing remotely from Windows or Linux:

1. Launch the setup from the location on the machine orchestrating the installation using the command above. This will open a prompt asking for the **Remote IBM i name/address**.
2. Enter the name/address of the target IBM i.
3. Login using the QSECOFR profile or an equivalent.

Step 3 Define the installation location.

**Example**

```
/HOME/...
```

Point to original installation location to use the same settings defined for the original installation (ports, installation location, service user).

Result The ARCAD Transformer Microservices Server is updated and available for use.

3.4.3 Uninstall

Follow the subsequent steps to uninstall the ARCAD Transformer Microservices Server on IBM i.

Step 1 Open a session on the IBM i where the server is installed, using the QSECOFR profile or an equivalent.

Step 2 Stop and delete the service using the following commands:

```
ADDLIB <AFSSTARTER LIBRARY NAME>  
ENDAFSSVR TFRMICROS  
DLTAFSSVR TFRMICROS DELETE(*YES)
```

Result The ARCAD Transformer Microservices Server is completely uninstalled.

4 Installing the ARCAD Transformer Microservices plug-in for RDi

There is only one ARCAD Transformer Microservices plug-in for RDi, which enables developers to:

- execute extraction analysis from piece of codes,
- externalize as ILE procedure piece of codes,
- manage and organize the ARCAD Microservices Projects and Rules, and
- execute Code Similarity searches.

4.1 Prerequisites

A RDi v9.6.11 (minimum) or v9.8.0.2 is necessary for a successful installation.



Important!

For any new installation the plug-in on RDi v.9.6, please refer to the Troubleshooting page to complete the installation.

4.2 Installing

ARCAD Transformer Microservices is an ARCAD extension plug-in that can be installed on RDi. Its installation requires the installation of plug-ins via two repositories:

- the content of the ARCAD Plug-in repository, and
- the content of the ARCAD Transformer Extensions repository.

Follow the subsequent steps to install the ARCAD Transformer Microservices plug-in(s), as well as the ARCAD Transformer Microservices plug-in.



Important!

To ensure a successful installation, the following the installation procedure needs to be repeated twice:

- during the first installation, in **Step 5**, set the path to the **ARCAD Plug-ins** repository to install the ARCAD Cores and ARCAD Solutions plug-ins, and
- during the second installation, in **Step 5**, set the path to the **ARCAD Transformer Extensions** repository to install the ARCAD Transformer Microservices plug-in.

Step 1 Download the installation package to your Local Site Repository. ARCAD delivers Eclipse plug-in packages in a .zip file. Save this .zip to your local environment. This location is the local site repository.

Note

The PC on which your Eclipse tool is installed must be able to access this location. It is recommended to save the installation files on the same PC or a shared network that the PC is connected to. There is no need to unzip this file after you save it to your local environment.

After installation, it is recommend that you keep this location. Eclipse will continue to recognize it, so when updates are available, you can copy them into this folder.

Step 2 From the **Help** menu of the Eclipse IDE, select **Install New Software**.

**Important!**

You must be logged in as an administrator.

Step 3 From the **Install** dialog, click the **Add** button.

Step 4 In the **Add Repository** dialog, enter a recognizable name, then click **Archive...** This creates a link to the Local Site Repository that you can come back to and that will be used for automatic updates.

Step 5 Navigate to and select the ARCAD plug-ins installation .zip file that you saved to your local environment. Click **OK**.

Result The **Location** field in the **Add Repository** dialog is set to the .zip file's path. You can add new versions to this path later to update.

Click **OK**.

Step 6 Select the plug-in(s) to install.

Click **Next >** to continue.

Step 7 Review the selected features in the **Install Details** dialog.

Click **Next >** to continue.

Step 8 Review and accept the license agreement

The **License Agreement** page presents the ARCAD license agreement for you to review. Please read it carefully.

When you have reviewed the agreement, select **I accept the terms of the license agreements**.

Step 9 Review and confirm the trusted certificates.

Click **OK**.

Step 10 Click the **Finish** button to launch the installation.

Step 11 Restart the workbench. Click **Yes** to restart.

Result ARCAD Transformer Microservices is installed and available for use.

4.3 Updating

Updating ARCAD plug-ins is similar to the original installation process. The prerequisites are the same and you must already have a previous version of the plug-in installed.

Note

When there is a major upgrade (as opposed to a simple cumulative patch) or a name change in one of the ARCAD plug-ins, then it is recommended to uninstall the current plug-in and restart your platform before installing the new plug-ins.

Upgrading ARCAD plug-ins only works correctly when the names of the installation files have not changed.

Note

If you try to re-install plug-ins instead of search for updates, as long as the correct .zip archive is selected, the plug-ins will update instead of reinstall. However, it is suggested to follow the following procedure to update plug-ins.

Follow the subsequent steps to update the ARCAD Transformer Microservices plug-in(s).

Step 1 Save the new version of the installation .zip in your Local Site repository (created when you first installed the plug-ins).

Step 2 From the **Help** menu of the Eclipse IDE, select **Check for Updates**.



Important!

You must be logged in as an administrator.

Step 3 The **Available Updates** dialog displays all of the plug-ins found in your update location that are out of date. Select the plug-in(s) to update.

Click **Next >** to continue.

Step 4 Review and confirm the updates.

Click **Next >** to continue.

Step 5 Review and accept the license agreement

The **License Agreement** page presents the ARCAD license agreement for you to review. Please read it carefully.

When you have reviewed the agreement, select **I accept the terms of the license agreements**.

Step 6 Review and confirm the trusted certificates.

Click **OK**.

Step 7 Click the **Finish** button to launch the installation.

Step 8 Restart the workbench. Click **Yes** to restart.

Result ARCAD Transformer Microservices is installed and available for use.

Note

Once the update is completed, the IBM i should be updated as well to avoid version compatibility issues. Each user must upgrade their plug-ins, however the upgrade to the server is only performed once.

4.4 Uninstalling

When there is a major upgrade (as opposed to a simple cumulative patch) or a name change in one of the ARCAD plug-ins, then it is recommended to uninstall the current plug-in(s) and restart your platform before installing the new plug-in(s).

Follow the subsequent steps to uninstall the ARCAD Transformer Microservices plug-in(s).

Step 1 From the **Help** menu of the Eclipse IDE, select **Installation Details**.

**Important!**

You must be logged in as an administrator.

Step 2 Click the **What is already installed** link in the lower right-hand corner of the **Install** dialog.

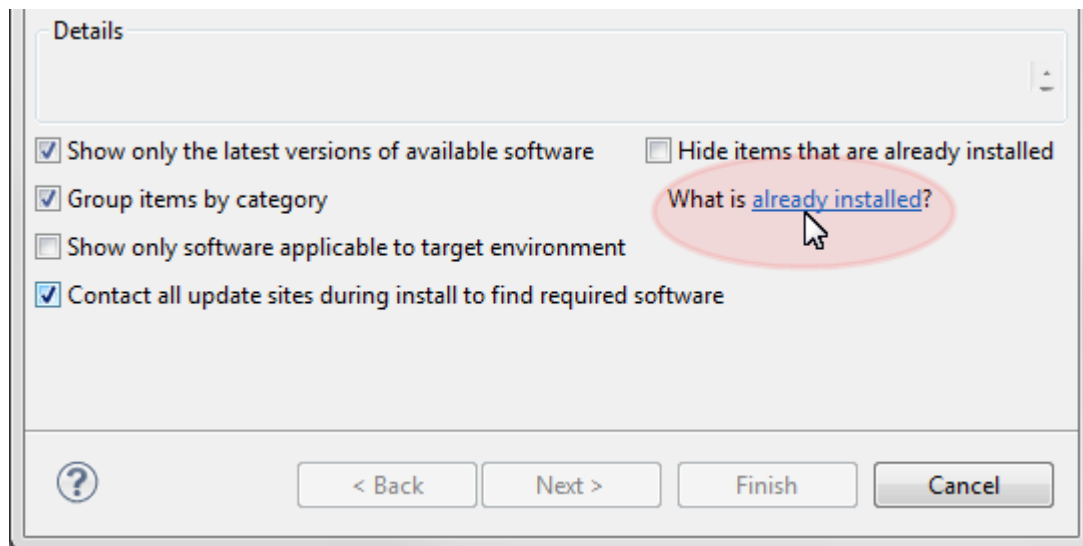


Figure 1: Already installed plug-ins

Step 3 Select the plug-in(s) to remove and click the **Uninstall** button.

Step 4 Click **Finish** to confirm.

Step 5 Restart the workbench.

Result The selected plug-in(s) is/are uninstalled.

5 Troubleshooting and FAQs for ARCAD Transformer Microservices

This page references the issues you can face during the installation of the ARCAD Transformer Microservices plug-in and it gives you some solution to solve them.

5.1 From RDi 9.6, ARCAD plug-ins installation failed with “Error reading signed content” exception for bcpkik (1.72.0), bcprov (1.72.0), bcutil (1.72.0) artifacts

Follow the subsequent steps to solve this issue.

Step 1 Copy the designated artifacts from the **plugins** folder located in the *ARCAD-Transformer-Extensions.zip*.

Step 2 Paste the artifacts in *C:\Program Field\IBM\SDP\plugins*.



Important!

It is not possible to define multiple aliases for the same user-defined type. The database does not allow it.

Step 3 Restart your RDi.

Step 4 Re-install, the ARCAD plug-ins for RDi after checking that this artifact is set as hidden items using **Hide items that are already installed**.

Result The ARCAD plug-ins installation is complete.